Page 1 | H5-A: The cut-off study summary

GreenWaste is a fictional training case. Use this study summary and the figures on page 2 as your source for today's questions.

§1 | Who and what determined participation?

GreenWaste offers equipment subsidies, installation help and training. The city file has **10,000 businesses**: 4,794 participant businesses and 5,206 comparison businesses (businesses that did not receive GreenWaste). Businesses with efficiency scores **58 or below participated**; those above 58 did not. Higher scores mean more efficient. The running variable is the score; **58 is the cut-off**. This rule is followed in the file, but it is not a lottery.

§2 | What does the fitted result compare?

The local model uses **545 businesses within two points of 58 (scores 56-60)**. The outcome is annual waste cost in AED per business, measured **12 months after**. Separate straight cost/score slopes are fitted on either side. The coefficient compares the fitted costs approaching 58 from the participating side versus the non-participating side. It differs from comparing average costs across all scores on each side, and is not the average gap across the whole city.

Result row	Coefficient AED	95% interval AED	Businesses
At/below cut-off: local jump	-721	-943 to -498	545

Cost 12 months after GreenWaste ~ below * distance; distance = score - 58. Separate linear slopes; window 56-60; HC2 robust interval. Reference: fitted cost above the cut-off at 58. The coefficient compares the fitted limits at the cut-off.

§3 | Which checks are supplied or missing?

A five-point window (scores 53-63) uses **1,260 businesses** and estimates **-784 AED**, interval **-934 to -634**. This sensitivity comparison does not prove causal validity. **Pre-programme manager age also jumps at 58: about -5.3 years in the five-point window.** It raises a concern about who is compared and other explanations for a cost jump. It does not prove a particular bias. No score-setting or manipulation diagnostic is supplied. The evaluator should investigate the scoring process, who could influence scores and other discontinuities. A causal reading requires comparable businesses immediately either side apart from programme access, without relevant sorting or other important jumps.

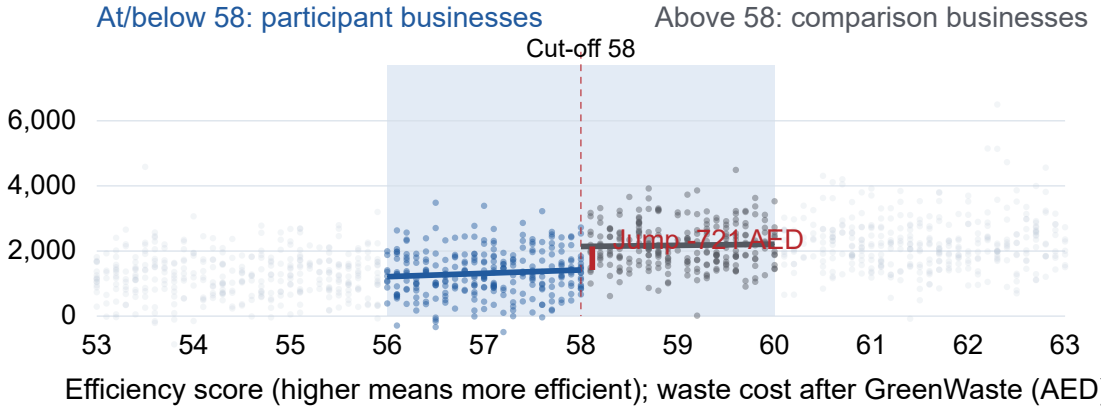
§4 | What follows for the decision?

On a positive-reduction scale, the local estimate is **721 AED**, interval **498-943**. Both are below the fictional **1,000 AED annual-saving minimum**, conditional on the model. Precision does not resolve the design concern. Most city participants are outside this local window. A result at the cut-off is not an estimated average effect for typical participants or the whole city. Full programme costs and national effects are not supplied.

Source: GreenWaste city data (10,000 businesses). Figures show records from the GreenWaste data and fitted lines, except the sorting illustration, which is hypothetical.

Page 2 | H5-A: visual reference

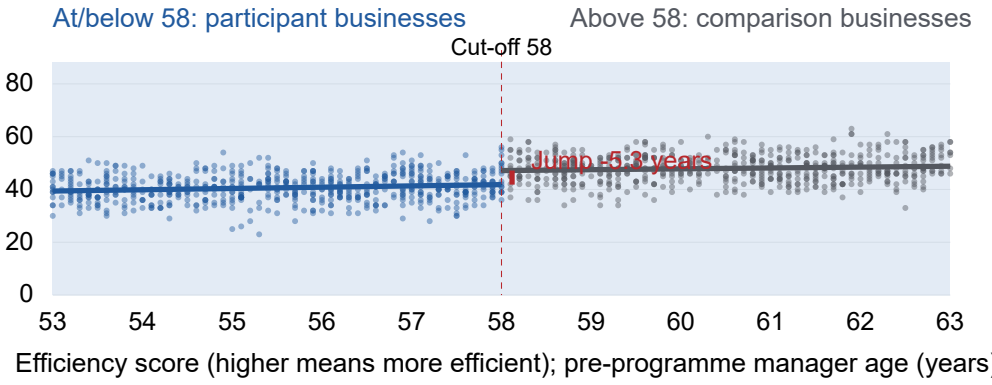
The fitted jump at 58: two-point window



Dots are city businesses in the GreenWaste data with scores 53-63. Shading marks scores 56-60 used in the main model. Lines approach the cut-off; the red gap is between their fitted limits.

Window	Businesses	Jump AED	95% interval AED
Within 2 points	545	-721	-943 to -498
Within 5 points	1,260	-784	-934 to -634

A pre-programme characteristic: manager age



This age discontinuity is a credibility concern, not a GreenWaste effect on age. Optional extension: adding age to the two-point model gives **-632 AED**, HC2 interval **-857 to -407**. Adjustment alone does not establish a valid local comparison.

Page 3 | Q1-Q2: visual comparison and interpretation

Q1 | What comparison should we make?

Use the plot on screen and the reference figure on page 2. Discuss with a partner.

(a) Identify the cut-off and the shaded window. Which side participated?

(b) Would comparing average costs on each side isolate the difference at 58? What else could affect that average gap?

(c) Follow the two fitted lines towards 58. What comparison do they make that the group averages do not?

Q2 | Read the fitted local jump

Use H5-A §2/table (page 1) and the fitted plot on page 2. Label the vertical gap at the cut-off. Which fitted cost is subtracted from which?

Interpret the coefficient: comparison, sign, units, timing and local population. Name what must be credible before calling it an effect.

Convert the negative interval to positive reduction. Where are the estimate and interval relative to 1,000 AED? What remains unresolved?

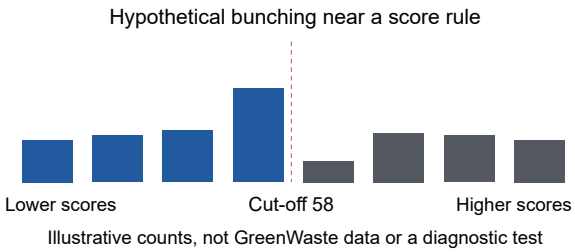
Page 4 | Q3: what would make this credible?

Use H5-A §1-4 (page 1) and page 2's figures. Mark **supported, concern or unknown**, citing evidence and a reason.

Check	Judgement, source and reason
Score rule followed	_____
Local comparison/window disclosed	_____
Other pre-programme differences	_____
Score setting/manipulation investigated	_____
Whole-city effect established	_____

What changes when the window widens? What does stability still not prove?

Why does the age jump matter? What would you ask the evaluator to investigate?



This is hypothetical, not a supplied manipulation test. Ask who set scores, when and whether businesses could influence which side they landed on.

Your independent reading and request

Keep H5-A open; close AI responses and peer comments. Write one supported local-result interpretation and one specific request. Explain how the answer could change your advice.

Page 5 | AI Snapshot: local result, wider claim

This response was written for the exercise. It is not a real AI answer. The prompt was: **Explain without jargon who this result applies to, and what a policy adviser should check before expansion.** Do not run this first prompt.

> The score threshold provides a regression discontinuity comparison. The fitted annual cost immediately below the cut-off is 721 AED lower than immediately above it, with a robust 95% interval of 498 to 943 AED lower. A wider five-point window also gives lower fitted costs, which is a useful sensitivity comparison.
>
> Businesses on either side of the threshold are effectively interchangeable because their scores are close. The manager-age difference is a background characteristic and does not affect this conclusion; adjusting for it would remove any remaining selection bias. The stable direction across windows confirms that the threshold isolates the programme effect.
>
> Implementation costs and practical capacity should still be checked. However, because most participants have scores below 58, this result describes the typical participant and supports citywide expansion. The interval is precise enough to show that the 1,000 AED saving requirement is met.

Underline support and cross out overclaims. Check **H5-A, pages 1-2**. Does a sensible caveat repair an unsupported assurance?

Improved prompt: run and compare

Explain H5-A without jargon to a policy adviser considering expansion. Businesses with a score at or below 58 took part; the local linear model uses scores 56-60 and 545 businesses, with separate slopes. The jump is -721 AED in annual waste cost 12 months after GreenWaste, HC2 interval -943 to -498; as a reduction this is 721, interval 498-943. Most city participants are outside this local window. Pre-programme manager age jumps by about -5.3 years in the five-point window. No score-setting/manipulation diagnostic is supplied. The fictional annual-saving minimum is 1,000 AED.

Ask essential clarifying questions first. Explain who the local comparison describes, what the coefficient and interval show, and what evidence is needed for a causal reading. Distinguish rule compliance from comparison validity. Request specific missing evidence; do not dismiss the age check, promise that adjustment fixes it, generalise citywide or claim that the saving minimum is met. Cite H5-A.

Copy this prompt and H5-A into **Microsoft Copilot**. Keep your AI good-practice card beside you. Page 6 is the prepared paper alternative.

Which claim improved? What still needs checking against the source?

Page 6 | Prepared response: paper alternative

Use this for the same task if Copilot is unavailable. This response was written for the exercise. It is not a real AI answer, and it is not additional study evidence.

> Before recommending expansion, I would ask how scores were assigned and recorded, whether businesses could influence them, and why manager age differs at the same threshold. I would also ask for other pre-programme checks and reasons for the chosen local window.
>
> H5-A compares fitted annual waste management costs 12 months after GreenWaste, immediately either side of score 58, using 545 businesses in scores 56-60 and separate slopes. It reports 721 AED lower cost on the participating side, with a model-based robust interval of 498 to 943 AED lower. A programme-effect reading requires a credible local comparison without other important jumps or sorting at the cut-off. The age discontinuity is a concern, not proof of a particular bias; controlling for age alone does not establish validity.
>
> A similar wider-window result is useful but does not settle these concerns. This is evidence near the cut-off, not an estimated average effect for all city participants. Both the estimate and interval are below the fictional 1,000 AED minimum, conditional on the model. I would investigate the design concern before relying on a causal interpretation, and request evidence for businesses further from the cut-off before extrapolating.

Check even a careful response

Match its comparison and numbers to H5-A §2 and page 2. Match design concerns and reach to §3-4. Does it distinguish recorded evidence, missing checks and an assumption?

Which assurance from page 5 has been removed? Is any wording still too strong or unclear?

Return to page 4 for your **own** interpretation and request, with AI responses closed.